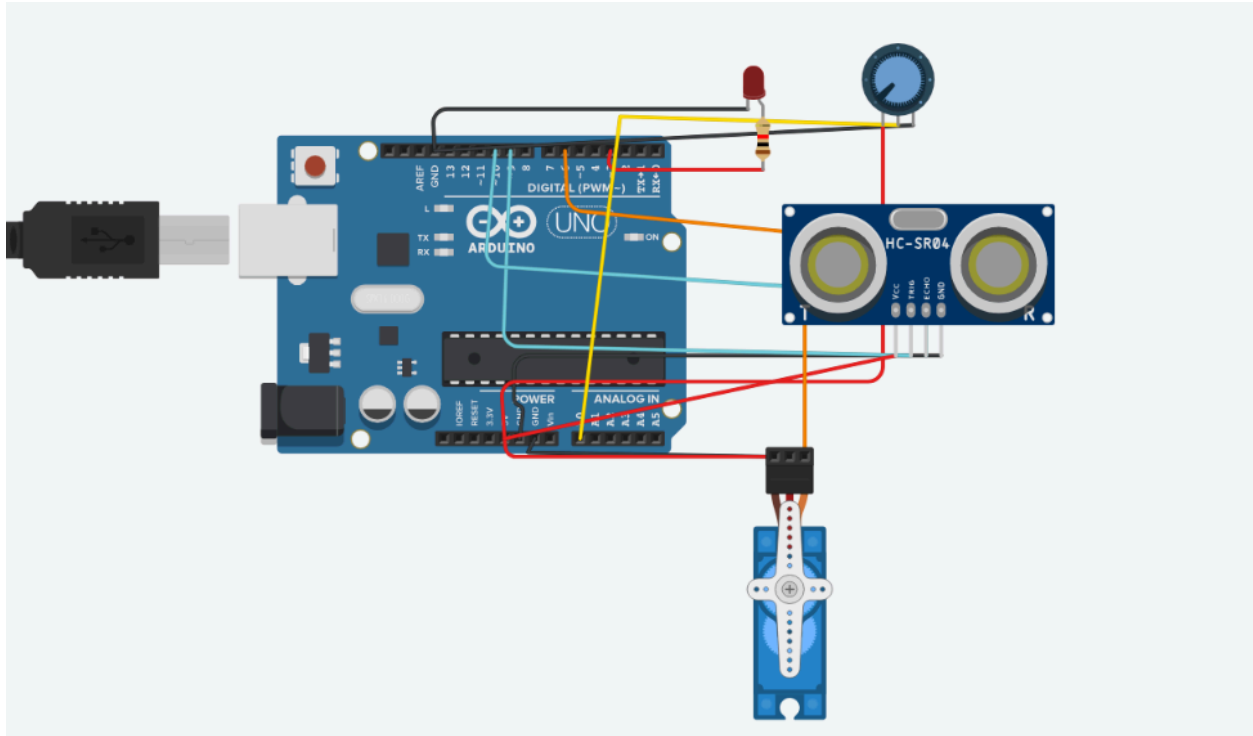




```
#include <Servo.h>
```

```
Servo gate;
```

```
int trigPin = 9;  
int echoPin = 10;  
int ledPin = 3;  
int potPin = A0;
```

```
long duration;  
int distance;  
int threshold;
```

```
void setup()  
{  
  pinMode(trigPin, OUTPUT);  
  pinMode(echoPin, INPUT);  
  pinMode(ledPin, OUTPUT);
```

```
  gate.attach(6);
```

```
  gate.write(0);  
  digitalWrite(ledPin, LOW);
```

```

    Serial.begin(9600);
}

void loop()
{

    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);

    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);

    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH);

    distance = duration * 0.034 / 2;

    threshold = map(analogRead(potPin), 0, 1023, 5, 50);

    Serial.print("Distance: ");
    Serial.print(distance);
    Serial.print(" cm  Threshold: ");
    Serial.println(threshold);

    if (distance <= threshold)
    {
        gate.write(90);          // Open gate
        digitalWrite(ledPin, HIGH); // LED ON
    }
    else
    {
        gate.write(0);           // Close gate
        digitalWrite(ledPin, LOW); // LED OFF
    }

    delay(100);
}

```